QQ-A-250/1F
30 June 1983
SUPERSEDING
QQ-A-250/1E
December 4, 1970

FEDERAL SPECIFICATION SHEET

ALUMINUM 1100, PLATE AND SHEET

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

The complete requirements for procuring the aluminum 1100 plate and sheet described herein shall consist of this document and the latest issue of the general specification QQ-A-250 (see 2.1).

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum 1100 plate and sheet.

1.2 Classification.

1.2.1 Tempers. The plate and sheet shall be of the following tempers: O, H12, H14, H16, H18, H22, H24, H26, H28, H112 or F, as specified (see 6.2). The definition of these tempers shall be as specified in the general specification QQ-A-250.

1.2.1.1 Temper substitutions. Material in either one of a pair of tempers (H12 and H22, H14 and H24, H16 and H26, or H18 and H28) may be supplied at the option of the producer or supplier, unless the supplying of one temper of the pair is expressly forbidden in the procurement documents (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specifications

QQ-A-250 - Aluminum and Aluminum Alloy, Plate and Sheet; General Specification for

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THIS DOCUMENT CONTAINS 12 PAGES.

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3. REQUIREMENTS

3.1 Chemical composition. The chemical composition shall conform to the requirements specified in table I.

TABLE I. Chemical composition 1/

Element	Percent	
	Minimum	Maximum
Iron plus silicon	--	0.95
Copper	0.05	0.20
Manganese	--	0.05
Zinc	--	0.10
Others, each	--	0.05
Others, total 2/	--	0.15
Aluminum	99.00	--

1/ Analysis shall routinely be made only for the elements specifically mentioned in table I. If, however, the presence of other elements is indicated or suspected in amounts greater than the specified limits, further analysis shall be made to determine that these elements are not present in excess of specified limits

2/ The sum of those "Others" metallic elements 0.010 percent or more each, expressed to the second decimal before determining the sum.

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3.2 Mechanical properties.

3.2.1 Mechanical properties of material as supplied. The mechanical properties parallel to the direction of the final rolling shall conform to the requirements specified in table II for the temper specified.

TABLE II. Mechanical properties

Temper	Thickness, inches	Ultimate tensile strength, ksi		Yield strength at 0.2 percent offset,	Elongation in 2 inches or 4 times D 2/, 3/,
		Minimum	Maximum	Minimum, ksi 1/	Minimum, percent
O	0.006-0.019	11.0	15.5	3.5	15
	0.020-0.031	11.0	15.5	3.5	20
	0.032-0.050	11.0	15.5	3.5	25
	0.051-0.249	11.0	15.5	3.5	30
	0.250-3.000	11.0	15.5	3.5	28
H12	0.017-0.019	14.0	19.0	11.0	3
	0.020-0.031	14.0	19.0	11.0	4
	0.032-0.050	14.0	19.0	11.0	6
	0.051-0.113	14.0	19.0	11.0	8
	0.114-0.499	14.0	19.0	11.0	9
	0.500-2.000	14.0	19.0	11.0	12
H14	0.009-0.012	16.0	21.0	14.0	1
	0.013-0.019	16.0	21.0	14.0	2
	0.020-0.031	16.0	21.0	14.0	3
	0.032-0.050	16.0	21.0	14.0	4
	0.051-0.113	16.0	21.0	14.0	5
	0.114-0.499	16.0	21.0	14.0	6
	0.500-1.000	16.0	21.0	14.0	10
H16	0.006-0.019	19.0	24.0	17.0	1
	0.020-0.031	19.0	24.0	17.0	2
	0.032-0.050	19.0	24.0	17.0	3
	0.051-0.162	19.0	24.0	17.0	4
H18 and H28	0.006-0.019	22.0	--	--	1
	0.020-0.031	22.0	--	--	2
	0.032-0.050	22.0	--	--	3
	0.051-0.128	22.0	--	--	4

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TABLE II. Mechanical properties (CON)

Temper	Thickness, inches	Ultimate tensile strength, ksi		Yield strength at 0.2 percent offset,	Elongation in 2 inches or 4 times D 2/, 3/,
		Minimum	Maximum	Minimum, ksi 1/	Minimum, percent
H112	0.250-0.499	13.0	---	7.0	9
	0.500-2.000	12.0	---	5.0	14
	2.001-3.000	11.5	---	4.0	20
H22	0.017-0.019	14.0	---	---	3
	0.020-0.031	14.0	---	---	4
	0.032-0.050	14.0	---	---	6
	0.051-0.113	14.0	---	---	8
	0.114-0.499	14.0	---	---	9
	0.500-2.000	14.0	---	---	12
H24	0.009-0.012	16.0	---	---	1
	0.013-0.019	16.0	---	---	2
	0.020-0.031	16.0	---	---	3
	0.032-0.050	16.0	---	---	4
	0.051-0.113	16.0	---	---	5
	0.114-0.499	16.0	---	---	6
	0.500-1.000	16.0	---	---	10
H26	0.006-0.019	19.0	---	---	1
	0.020-0.031	19.0	---	---	2
	0.032-0.050	19.0	---	---	3
	0.051-0.162	19.0	---	---	4
F	All	4/	4/	4/	4/

1/ 1,000 ksi equals 1,000 psi
2/ Not required for material 0.500 inch or less in width
3/ "D" is specimen diameter
4/ No requirements

3.2.2 Bend properties. Bend-test specimens taken from material shall be capable of withstanding, without cracking, the bend test specified in the general specification QQ-A-250. The values for bend factor "N" are listed in table III.

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TABLE III. Bend test factor "N"

Thickness of material, inches	Temper			
	O	H12 or H22	H14 or H24	H16 or H26
0.006-0.008	0	--	--	4
0.009-0.016	0	--	0	4
0.017-0.162	0	0	0	4
0.163-1.000	0	0	0	--
1.001-2.000	0	0	--	--
2.001-3.000	0	--	--	--

4. QUALITY ASSURANCE PROVISIONS (see the general specification QQ-A-250)

4.1 Sampling. Sampling for conformance to the requirements of section 3 shall be as specified in the general specification QQ-A-250).

5. PAREPARATION FOR DELIVERY (see the general specification QQ-A-250)

6. NOTES

6.1 Intended use. Aluminum 1100 plate and sheet are intended for use where very good formability and good resistance to corrosion are required and when relatively high strength is not required. The material is readily welded.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number and date of this specification
(b) Form, dimensions, quantity and temper required (see 1.2.1)
(c) If substitution for temper ordered is forbidden (see 1.2.1.1)
(d) Requirements for sizes not specifically covered by this detail specification (see the general specification QQ-A-250)
(e) Special end use requirements
(f) Selection of applicable levels of preservation/packaging and packing required, if other than level C (see the general specification QQ-A-250)

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MILITARY INTEREST:

Custodians
Army-MR
Navy-AS
Air Force-20

Review Activities
Army-CR, MI
Navy-OS, SH
Air Force-85

User Activities
Army-ME

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
NASA-JFK, MSF
DOE-BPA

PREPARING ACTIVITY:

NAVY-AS

DOD project 9535-0240

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